ResQ-MARL: PROJECT PIVOT MASTER PLAN

From Entertainment Game to Autonomous Disaster Response & USAR Testbed

Scope: 3-Person Team Division, 3-Phase Timeline, and Feature Inventory | **Status:** Official Execution Plan

> ***The Pivot Mission:*** *Repositioning our Multi-Agent Reinforcement Learning engine into* ***ResQ-MARL****, an interactive simulation and digital twin platform that trains autonomous robotic swarms (UAVs/UGVs) to navigate collapsed earthquake rubble, detect trapped victims via simulated thermal sensors, clear dynamic debris, and escort survivors to a designated Safe Triage Zone. Enables human incident commanders to survey sites and clear debris using* ***VR and touchless webcam hand tracking****.*

1. Work Breakdown: 3-Person Team Division

Responsibilities are partitioned into 3 independent streams so all members can develop simultaneously:

Team Member	Key Technical Responsibilities	Target Deliverables
Person 1: Backend, AI & Simulation Lead	• Implement Safe Triage Zone trigger & extraction routing. • Refactor capture rewards into rescue & survival milestone rewards. • Build automated 10-trial batch test runner in GenerationComparer.cs.	Safe Triage Zone logic, survivor extraction state loop, and batch benchmark harness.
Person 2: Spatial Analytics & Hand Tracking	• Build 30x30 thermal/LiDAR floor heatmap (SpatialThermalHeatmap.cs). • Develop Python MediaPipe webcam hand-tracking script. • Implement Unity UDP gesture receiver for touchless debris movement.	Live thermal coverage heatmap (key 'H') and touchless webcam hand controller.
Person 3: UI, Presets & Mission Exporter	• Reskin HUD into Emergency Incident Command Post. • Create 3 disaster blueprints (School, Corridor Chokepoint, Rubble Field). • Build one-click CSV mission telemetry exporter.	Disaster Incident HUD, 3 realistic collapsed presets, and CSV mission data exporter.

2. Three-Phase Execution Roadmap

Phase 1: Logic & Triage Loop (Days 1–2): Add the green Safe Triage Zone to the arena floor, update GameController.cs to tag victims as 'Rescued' and guide them to safety, and ensure ML-Agents models continue running smoothly.

Phase 2: Heatmap, Gestures & Presets (Days 3–4): Implement the 30x30 thermal sensor floor heatmap (toggleable with key 'H'). Deploy the Python webcam hand-tracking bridge to allow pinch-based debris manipulation. Configure the 3 collapsed building presets.

Phase 3: Visual Polish & Defense Rehearsal (Day 5): Reskin UI into an Emergency Command Post, connect one-click CSV mission telemetry export (logging MTTR and sweep clearance rate), and rehearse the 5-minute review presentation.

3. Feature Inventory: What to Add vs. What to Remove/Rename

Category	Item	Action / Disaster Transformation
New Feature	Safe Triage Zone	Designated recovery base where rescued victims are guided and secured.
New Feature	Thermal Floor Heatmap	Floor grid highlighting scanned rooms vs dangerous unexplored voids ('H' key).
New Feature	Webcam Hand Controller	Touchless hand gestures to pinch, lift, and clear virtual rubble via MediaPipe.
New Feature	Disaster Presets	Instant loading of collapsed school, blocked corridor, and rubble scenarios.
To Remove / Rename	'Hide and Seek' Labels	Replace with 'Urban Search and Rescue (USAR) Simulation & Digital Twin'.
To Remove / Rename	'Hiders' & 'Seekers'	Rename to 'Trapped Survivors' and 'Autonomous Rescue Swarm (UAVs/UGVs)'.
To Remove / Rename	'Capture' / 'Demolish'	Rename to 'Survivor Secured' and 'Heavy Excavation / Debris Removal'.

4. Recommended Visual & Aesthetic Changes

- **Rescue Drones (Seekers):** High-visibility hazard amber body rings with warm orange/amber thermal vision cones.
- **Trapped Survivors (Hiders):** Pulsing cyan life-sign rings with overhead status badges: [TRAPPED] -> [EXTRACTING] -> [SAFE].
- **Debris Blocks:** Retextured with cracked concrete, broken masonry, and hazard warning textures.
- **Safe Triage Base:** Illuminated green perimeter box with an emergency medical/hazard symbol on the floor.
- **Incident Command HUD:** High-contrast emergency response theme (Tactical Slate, Amber, Green) displaying real-time MTTR and survivor triage counters.